

MaxFilter Guidelines

These instructions are to give a rough idea of the MaxFilter usage and reasonable parameter ranges. Please note that there may be some errors in these guidelines.

Detailed instructions how to use of MaxFilter are available in the *Elekta Neuromag MaxFilter User's Guide*.

Abbreviations:

- IAS Internal Active Shielding, interference suppression method using subsets of magnetometers and compensation coils inside the magnetically shielded room.
- EAS External Active Shielding, stand-alone system with a fluxgate sensor and compensation coils outside the magnetically shielded room.
- cHPI continuous Head Position Indicator measurement, used online to activate continuous head tracking in MEG measurements
- SSP Signal Space Projection, mathematical algorithm to attenuate ambient noise in MEG recordings, used typically online. Based on the empty room data, site-specific upright60, upright68, and supine SSP vectors.
- SSS Signal Space Separation, *Basic SSS*, mathematical algorithm to attenuate noise in MEG recordings offline, uses factory-set system specific calibration /neuro/databases/sss/sss_cal.dat and crosstalk correlation /neuro/databases/ctc/ct_sparse.fif files.
- tSSS Temporal extension of SSS algorithm, *Spatiotemporal tSSS*.

General difference between SSS and tSSS

SSS is a spatial filter which effective in attenuating magnetic artifacts that are not very close to the sensor array whereas tSSS is a spatiotemporal filter which is more effective in attenuating magnetic artifacts close to the sensor array, i.e., on the head, chest, and clothes or stimulator artifacts in the close vicinity of the sensor array. tSSS uses the correlation between the internal (inside the sensor array), intermediate (at the sensor array), and external (outside the sensor array) signal compartments to select and attenuate those signals that are highly correlated in these compartments.

Whenever the original data is being processed with MaxFilter, a new data file will be generated and the original file will be intact.

tSSS method should be applied on the raw data since it is far more effective when applied on the raw data.

The original SSP vectors will not be available in the processed data files since they are obsolete after SSS or tSSS operations.

MaxFilter options

Multipole extensions: defaults are recommended (Inside 8, outside 3).

Component selection: Default *optimize inside basis* is recommended.

Bad channels: Use automatic bad channel scanning (scan all raw data buffers) or add bad channels manually. Bad channels not defined correctly may cause a spreading of artefacts to the neighbouring channels. The original data file should be carefully inspected prior to maxfiltering, *e.g.*, with graph compare setup.

NOTE: cHPI comes available during the recording after accepting the initial HPI measurement. Should the cHPI used click it on before clicking raw data recording on. Should one or a few MEG channels are not working correctly it is recommended to record all MEG channels during the recording.

Active shielding

If no active shielding (IAS/EAS) has been used during the MEG recording, either SSS or tSSS can be used off-line, if needed, to attenuate artifacts in the recordings.

If EAS has been applied, either SSS or tSSS can be used off-line, if needed, to attenuate artifacts in the recordings.

If IAS has been used, SSS/tSSS has to be applied to the data the data to get rid of the IAS effects on the MEG data.

Parameters for tSSS

Length of raw data buffer (Default 10 s), reasonable range 10 s or higher. Using lower values may cause artifacts in MEG data.

Subspace correlation limit (default 0.98), reasonable range 0.8–0.98; depends on the artifact type and distance of the artifact. Using too low correlation limit may cause attenuation of brain signals.

cHPI considerations

If the cHPI has been used during the recording use SSS/tSSS before analysing the data. Some residual HPI coil signals may be visible after the SSS/tSSS filtering. The residual coil signal can be removed by two ways, either you use a low-pass filter when applying SSS/tSSS (Use lowpass filtering enabled and corner frequency below the coil frequencies; recommended) or use additional low-pass filter in source modelling/plotting software.

Off-line averaging

Once the raw data has been processed with MaxFilter it can be averaged off-line using the MaxFilter's Averager module. The Averager will use the averaging categories and rejection settings used in the original data acquisition session. If needed, the rejection settings can be adjusted from the Averager Rejection Limits.